Cancellation, Observable Memory, and Exact Causal Seeds

A Finite Probabilistic Bridge for a Moment Writer

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Abstract

We give a finite probabilistic realization of the sum-and-moment writer used in ONE. For passive adaptive observations that include the running sum at every step, the optimal exact causal seed entropy equals the joint entropy of all deposits. We also give a terminal cancellation example whose seed cost is zero for sum-only observation and equals the deposit entropy when the first moment is observable. The results follow directly from deterministic entropy monotonicity and exact reconstruction. No claim of originality or physiological interpretation is made.

1 Model and simulation requirement

Fix $n \geq 1$ and deterministic real addresses $x_1, \ldots, x_n$. Let $Z = (Z_1, \ldots, Z_n)$ be a complex-valued random vector whose joint law μ has finite support. No independence assumption is made. Define the writer by

$$S_0 = M_0 = 0, \qquad S_t = S_{t-1} + Z_t, \qquad M_t = M_{t-1} + x_t Z_t, \quad 1 \leq t \leq n.$$

These are the sum-and-moment updates considered in ONE, with $x_t = \log p_t$ as one possible address choice [1]. The probability law μ is an additional modeling input; it is not implied by the deterministic update identities.

At step t, an observer selects $A_t \in \{\mathsf{S}, \mathsf{M}\}$ and receives $Y_t = S_t$ or $Y_t = M_t$, respectively. Selection is by a deterministic adaptive policy using the earlier actions and answers. Observation is passive: it changes neither the deposit vector nor the writer dynamics. The distribution μ is the same under every policy. Finite support makes every time-specific output alphabet finite, even though its labels are complex numbers.

An exact causal simulator consists of a finite seed U with a policy-independent law and one deterministic decoder

$$Y_t = D(U, H_{t-1}, A_t), \qquad H_{t-1} = (A_1, Y_1, \ldots, A_{t-1}, Y_{t-1}).$$

It must reproduce the target transcript law for every policy. The decoder is a single function: identical histories and actions receive identical answers from a given seed atom. Define

$$C_n = \inf\{\mathrm{H}(U) : U, D \text{ form an exact causal simulator}\}.$$

All entropies are Shannon entropies in bits.

2 Exact optimal cost

Theorem 1 (Observable writer)**.** *The minimum is attained, and*

$$\boxed{C_n = \mathrm{H}(Z_1, \ldots, Z_n).}$$

Proof. For the upper bound choose $U = Z$ with law μ. Given U, compute all partial sums and moments. At each history return the readout requested by the current action. This uses one decoder for all policies and satisfies shared-history consistency. Its transcript is exactly the transcript obtained from the writer with deposits Z. Hence $C_n \le \mathrm{H}(Z)$.

For the lower bound use the admissible policy that always selects S. Its output is $(S_1, \dots, S_n)$, from which

$$Z_t = S_t - S_{t-1}, \qquad S_0 = 0.$$

The deposit vector and this output vector determine one another. Therefore $\mathrm{H}(S_1, \dots, S_n) = \mathrm{H}(Z)$. In every exact simulator, the output under this fixed policy is a deterministic function of the seed. Thus $\mathrm{H}(Z) = \mathrm{H}(S_1, \dots, S_n) \le \mathrm{H}(U)$. Combining both bounds proves the identity and attainment. □

Corollary 1 (Dependent and independent deposits)**.** *In general,*

$$C_n = \sum_{t=1}^{n} \mathrm{H}(Z_t \mid Z_1, \dots, Z_{t-1}).$$

If the deposits are independent with common law P, then $C_n = n\mathrm{H}(P)$.

Proof. Apply the entropy chain rule and, in the independent case, independence. □

The cost is the entropy of an initially sampled seed. It does not require fresh randomness to be physically supplied at each step: after sampling, the decoder can run deterministically. It is not a bound on energy, execution time, or the memory needed to represent exact numerical labels.

3 Cancellation with observable memory

Now consider a different interface: only a terminal readout after two deposits, with no earlier observations. Choose $x_1 \ne x_2$ and a finite-valued complex random variable W, and set

$$Z_1 = W, \qquad Z_2 = -W.$$

Then

$$S_2 = 0, \qquad M_2 = (x_1 - x_2)W.$$

Theorem 2 (Readout-dependent cost)**.** *For the sum-only terminal interface the optimal seed entropy is zero. For the terminal interface permitting the choice of either sum or moment, the optimal seed entropy is* $\mathrm{H}(W)$*:*

$$\boxed{C_{\mathsf{S}} = 0, \qquad C_{\{\mathsf{S},\mathsf{M}\}} = \mathrm{H}(W).}$$

Proof. The terminal sum is identically zero, so a constant seed suffices. For the larger interface, the seed $U = W$ generates both readouts. Conversely, the policy selecting the moment receives $(x_1 - x_2)W$. Since multiplication by the nonzero scalar $x_1 - x_2$ is injective, this output has entropy $\mathrm{H}(W)$. Deterministic entropy monotonicity gives the matching lower bound. □

This comparison concerns a change in the observation interface, not a change in the writer or its law. In particular, if S_1 were observable, it would already reveal W; the zero-cost statement would no longer apply to that enlarged interface.

The example proves that a constant readout can coexist with arbitrarily large finite information accessible through another readout: take W uniform on any N distinct values, obtaining $\mathrm{H}(W) = \log_2 N$. It does not assert that all silent readouts conceal information.

4 Connection to controlled-tree simulation

For any policy π, the deposit vector determines its full transcript: there is a deterministic map F_π with target law $(F_\pi)_{\#}\mu$. Conditional probabilities of successive answers on reachable histories define a finite controlled tree. The explicit seed construction above realizes all these laws using a single history-consistent decoder.

In the positive finite-tree setting, the companion causal Shannon theorem [2] gives

$$L(T) \le C(T) \le L(T) + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2},$$

where $L(T)$ is the mean lower envelope of policy transcript information distributions. This general theorem is contextual, not a premise of the two direct proofs above. Those proofs also allow zero-probability deposit configurations and deterministic readouts. No general extension of the companion theorem to zero branches is claimed here.

The exact identity for the observable writer follows from a stronger feature of this special experiment: one admissible policy reconstructs the entire deposit vector. It must not be extrapolated to arbitrary controlled systems, especially systems in which interventions modify the dynamics.

5 Interpretation and limits

The bridge has four explicit ingredients:

writer updates $\longrightarrow$ deposit law $\longrightarrow$ adaptive observation interface $\longrightarrow$ optimal seed entropy.

Only the first ingredient is supplied by the deterministic sum-and-moment identities alone. An exact stochastic realization requires the other ingredients to be specified.

For the study of stillness, the relevant mathematical lesson is limited but precise: absence of variation in one report does not determine the information available through other measurements. Identifying this writer with a biological process would require independently justified measurement maps and experimental probability laws. No claim about quantum consciousness, subjective experience, or physical energy consumption follows from these entropy identities.

References

[1] S. L. Onweller, *ONE*, supplied edition `ONE_E24_Gabriels_Horn_Integrated_Pass7.pdf` (2026), Theorem PS.1, printed p. 17. `https://doi.org/10.5281/zenodo.23575568`. The present note uses the displayed writer updates, not an audit or endorsement of the full monograph.

[2] J. Mikulík, *Finite Shannon Theorem*, companion development. `https://github.com/jendamikulik/finite-Shannon-theorem`. The repository is cited for context; its current formalization is not independently checked in this note.